

Curriculum Vitae

1. Key Qualification:

Drs. W.D. Langeraar (MSc) is a distinguished senior GIS and Mapping Specialist renowned for turning intricate data into powerful visual narratives. With a professional footprint spanning multiple continents and disciplines, he specializes in creating the "information bridge" that turns spatial analysis into strategic success. He doesn't just map data; he engineers the clarity required for world-class decision-making.



Wine Langeraar in Dushanbe, Tajikistan, 30 Jan 2026

2. Name:

Wijnand ("Wine") Durk LANGERAAR

3. Coordinates:

Netherlands address: Herenweg 129, 3625AD Breukeleveen
Indonesia address: Jalan Pengembak 36-C, 80228 Sanur Bali
Email: winedl@gmail.com
Dutch mobile number: +316 2698 8005
Indo Mobile and WhatsApp number: +62 821 2806 1534

4. Nationality:

Netherlands

5. Date of Birth:

24 August 1948

6. Education:

- B.Sc. geology, University of Groningen, Netherlands, 1973.
- M.Sc. degree general geology, with major subjects: (1) Sedimentology (Thesis: Palaeo-environment of an Eocene Sedimentary Basin, Pyrenees, Spain), (2) Mining Politics and Mining Law (Thesis: The Oil Politics of Libya), *cum laude*¹, Universities of Utrecht, Delft and Leiden, Netherlands, 1977.

7. Membership of Professional Associations:

Dutch Remote Sensing Society, LinkedIn

8. Other Training:

- Aerial Photo Interpretation Applied to Soil Survey (with emphasis on remote sensing), International Institute for Geo-information Science and Earth Observation ITC, (now Technical University Twente), Enschedé, Netherlands, 1977
- Arabic speaking and writing, at FAO, Rome, 1979
- Italian conversation, at FAO, Rome, 1979
- Project Management course, MDF, Ede, Netherlands, 1986
- Advanced Esri® ArcGIS training, NGIS, Perth WA, Australia, May 2006
- GeoSamba training in web-enabled GIS, NGIS, Perth WA, Australia, March 2010
- Online course Python scripting for ArcGIS for Desktop ver. 10, January 2013
- Web GIS webinar by Esri Australia, August 2022

9. Countries of Work Experience: (Last 20 years)

Afghanistan, ADB Central and West Asia countries, Fiji, Gabon, Indonesia, Iraq, Iraqi Kurdistan, Kyrgyzstan, Mozambique, Mongolia, Myanmar, Pakistan, Papua New Guinea, Philippines, Rwanda, Tajikistan, Timor-Leste, United Arab Emirates, Vanuatu, Viet Nam

10. Computer literacy:

Proficient in the use of, and in possession of, official user licenses for the following computer programs:

- Esri®: ArcGIS ArcMap ver. 10.8.2 and ArcGIS Pro ver. 3.7.0
- Blue Marble®: Global Mapper ver. 27
- Clark Labs®, Clark University: TerrSet
- Open-source GIS programs (QGIS, SAGA-9.5.0, Google Maps/Earth/Earth Engine, etc.)
- All business software

11. Languages:

Speaking

Reading

Writing

¹Notified copy of diploma attached

Dutch
English
French
Indonesian (Malay)
Italian
German
Spanish
Arabic

Mother tongue
Fluent
Fluent
Fluent
Good
Good
Fair
Basic

Mother tongue
Fluent
Fluent
Fluent
Good
Fair
Fair
None

Mother tongue
Fluent
Fluent
Fluent
Fair
Fair
Fair
None

12. Employment Record:

(Last 25 years)

From: June 2026
Name of assignment or project:

Employer:

Client:

Positions held:
Main Project Features:

Activities performed:

To: July 2026
Sustainable Highlands Highway Investment Program (SHHIP)
Tranche 2
Bonifica GROUP. Via della Camilluccia, 67
001835 Rome, ITALY
Government of **Papua New Guinea**, funded by ADB Loan
4094/4095-PNG and OPEC Fund Loan No. 14606P, PNG
GIS & Mapping Specialist
Culverts and bridges design work along Highlands Highway around
Goroka and surrounding areas, PNG
Working from home, ascertained catchments upstream of 51
culverts along the Highlands Highway and generated information
on area size, average slope, elevation, and any other information
needed by the Project's hydrologist.

From: November 2025
Name of assignment or project:

Employer:

Client:

Positions held:
Main Project Features:

Activities performed (in process):

To: present (Intermittent)
TA to develop a health facilities network master plan to optimize
health services delivery
Ginger SOFRECO, 92-98 boulevard Victor Hugo, 92115 Clichy
Cedex, France
Ministry of Health and Social Protection of Population of the
Republic of **Tajikistan** - World Bank funded
Geospatial Planning Specialist
Improve the quality and efficiency of primary healthcare services
(PHC) in selected districts and strengthen the national capacity to
respond to public health emergencies.
Working both on-site in Tajikistan and from home base in Bali,
Indonesia:
(i) In process of mapping and geo-referencing the complete
health facility network (PHC, secondary, tertiary)
(ii) Conducting rigorous spatial analysis of geographic
accessibility, travel time, seasonal constraints, and disaster
exposure of medical facilities
(iii) Engineering of interactive GIS-based maps underpinning
catchment area delineation, patient referral pathways, and
investment prioritization
(iv) Enabling evidence-based, data-driven decision-making on
facility location, consolidation, and upgrading.
(v) Delivering targeted capacity-building programmes to assigned
Ministry staff using Quantum GIS

From: March 2023
Name of assignment or project:

Employer:

Client:

Positions held:
Main Project Features:

To: October 2025 (Intermittent)
Central Asia Regional Economic Cooperation Corridors 1 and 3
Connector Road Project (Phase 2), Institutionalization of RAMS in
Ministry of Transport and Communications OTC's Business Process
Roughton International Limited. A2 Omega Park, Electron Way,
Chandlers Ford, Hampshire SO53 4SE United Kingdom
Ministry of Transport and Communications
of the **Kyrgyz Republic** - ADB funded
Road Database Specialist
Improving the sustainability of the Kyrgyz road network by
implementing, populating and operationalizing a Road Asset
Management System (RAMS). The RAMS will form the basis for
planning and budgeting for the road network in the Kyrgyz
Republic.

Activities performed:

Working onsite in Kyrgyzstan, and from home-base in Bali:

- (i) Various GIS activities, such as mapping Kyrgyzstan's road network, analyzing and determining landslide-prone stretches of roads, calculating percentages of road elevation, production of shaded relief background maps for online information system, etc.
- (ii) Update of the existing RAMS Database and Maintenance Planning System
- (iii) Support preparations of Three-Year Rolling Plan, and RAMS Planning and Operation Manuals.

From: December 2020

To: March 2023 (intermittent)

Name of assignment or project:

Detailed Engineering Design of Bataan-Cavite Interlink Bridge (BCIB) Project

Employer:

RENARDET S.A., rue Rothschild No. 50, Genève, Switzerland, in association with T. Y. Lin International, 2 Harrison St, 94105 San Francisco, USA and Pyunghwa Engineering Consultants Ltd. (PEC), 454 Gwanak-daero, Dongan-gu Anyang, 431810 South Korea

Client:

Department of Public Works and Highways (DPWH), Government of **Philippines**, and funded by the Asian Development Bank (ADB Loan 3886-PHI): Infrastructure Preparation and Innovation Facility (IPIF)-Additional Financing

Positions held:

GIS/Survey Lead

Main Project Features:

Consulting Services for the Detailed Engineering Design of Bataan-Cavite Interlink Bridge Project under Infrastructure Preparation and Innovation Facility (IPIF)-Additional Financing ADB Loan No. 3886-PHI. The project comprises a marine bridge of about 26km and about 6km of onshore viaducts and roads. The marine bridge will include marine viaducts and two cable-stayed bridges of 1800m in the South Navigation Channel and 800m at the North Navigation Channel, to allow for marine traffic.

Activities performed:

Working from home bases in the Netherlands and Bali, Indonesia:

- (i) Production of extensive map series for BCIB Environmental Impact Assessment activities
- (ii) Leading GIS/Survey team bringing together all sub-contractor results and other GIS inputs,
- (iii) Mapping and monitoring of sub-contractor progress
- (iv) Producing GIS Technical Notes on a variety of *Capita Selecta* subjects with bearing on the bridge project's land and sea areas
- (v) Monitoring NAVPHIL announcements, maintaining and updating a potential maritime threats map of the project area.

From: November 2020

To: January 2022 (Intermittent)

Name of assignment or project:

East Bali Poverty Project

Employer:

idem

Client:

idem

Positions held:

GIS and Mapping Specialist

Main Project Features:

The East Bali Poverty Project (EBPP) is a non-profit organisation helping the thousands of people living in abject poverty without water, sanitation, roads, schools, health facilities and electricity in East Bali, **Indonesia**

Activities performed:

Working from home base in the Netherlands (*Pro Deo*):

- (i) Engineered detailed topographic maps for the project area leveraging Indonesia's high-resolution DEMNAS digital elevation model data, against a background of ArcGIS Online WorldImagery, and
- (ii) Remotely coached project staff in the use of GPS tracking, and used the resulting GPS track and waypoint information in map updates.

From: September 2020

To: November 2020

Name of assignment or project:

Contract V.O. #4 for Detailed Design of 15 Road Failure Sections under CW2 CONTRACT (CSTB 3775) of the Highlands Highway.

Employer:

RENARDET S.A., rue Rothschild No. 50, Genève, Switzerland.

Client:	Department of Works, Goroka, Papua New Guinea , funded by the Asian Development Bank
Positions held:	International GIS Specialist
Main Project Features:	Provision of physical and demographic information – including extreme rainfall events that occurred in the last 20 years - for basins upstream of critical sections along the Highland Highway, and delivery of other information needed to allow optimum design of culverts at stream crossings and alternative road corridors.
Activities performed:	Worked from home base in the Netherlands: (i) Using NASA IMERG data provided 2001 to 2020 rainfall data: all annual, selected monthly, daily, and up to half-hourly rainfall intensity for days with extreme precipitation intensities for catchments upstream of critical section of the Highlands Highway, PNG. (ii) Assessed pertinent parameters of basins such as surface area, monthly average rainfall, basin area population density, maximum stream length and basin width, lowest and highest point in basin, mean slope of the basin, forest cover.
From: April 2020	To: September 2020
Name of assignment or project:	Socio-Economic Impact Evaluation and Planning and Prioritisation Toolkit for Low Volume Rural Roads in Afghanistan (AFG2159A)
Employer:	PLANET S.A., Apollo Tower, 64, Louise Riencourt Street 115 23, Athens, Greece
Client:	Ministry of Rural Development and Rehabilitation (MRRD), Afghanistan , administrated by ReCAP Project Management Unit, Cardno Emerging Market (UK) Ltd, Clarendon Business Centre, Level 5, 42 Upper Berkeley Street, Marylebone, London W1H 5PW, and funded by the UK Government
Positions held:	GIS Specialist
Main Project Features:	AFG2159A has two components – the first component involves an analysis of the socio-economic impact evaluation of rural roads in Afghanistan; and the second component involves the design and development of a planning and prioritisation toolkit. Contributed to the second component.
Activities performed:	Worked from home base in the Netherlands. Deployed geospatial data to generate multi-layered maps from online open sources and ArcGIS Online WorldImagery, including: (i) Extracting a number of physical road parameters (road alignments, surface material, grade/steepness, road width in metres, length, connections, environment traversed by the road, etc.), and (ii) Generating additional information, such as population distribution and density of the areas traversed by the roads (obtained from WorldPop), number of households per village and per district, average elevation, and access to water.
From: January 2020	To: April 2020
Name of assignment or project:	Detailed Design of the Remaining Sections of Highlands Highway under RIC (CSTB 3776); MANGIRO - MIUNDE SECTION (SIMBU PROVINCE), PNG
Employer:	RENARDET S.A., rue Rothschild No. 50, Genève, Switzerland.
Client:	Department of Works, Goroka, Papua New Guinea . Funded by the Asian Development Bank
Positions held:	GIS Specialist
Main Project Features:	Assisted with obtaining (i) rainfall data records, and (ii) maximum annual rainfall in Simbu Province since 2001 till present – and other data needed to allow optimum design of culverts at stream crossings.

Activities performed:

Worked from home bases in Bali and the Netherlands:

- (i) Using NASA IMERG data, provided 2001 to 2019 all annual, selected monthly and daily rainfall data, with corresponding stream flow calculations, for catchments upstream of critical culverts.
- (ii) Using hi-res Esri WorldImagery, SRTM elevation data, and WorldPop population data, assessed pertinent parameters for basins such as surface area, number of people living in basin area with population density, maximum stream length, lowest and highest point in basin, maximum width, mean slope of the basin, vegetation cover.

From: September 2019

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: December 2019

Reducing deforestation through improved spatial planning in Papua provinces

LAND EQUITY INTERNATIONAL Suite 12, 74 Kembla St
WOLLONGONG NSW 2500 PO Box 798

WOLLONGONG NSW 2520 Australia

DFID / UK Government: **Indonesia**

Geospatial Data Lead

The Government of Indonesia has set ambitious targets for curbing deforestation and peatland degradation but has faced difficulties in meeting these targets. The project has thus three outputs: (i) Improved revision and implementation of Papua and West Papua spatial plan, (ii) Improved transparency and enhanced constituency in spatial planning at the provincial level, and (iii) National policy dialogues and engagement to support Papua's commitment to protect its forests.

Activities performed:

Working from home base in the Netherlands:

- (i) Downloaded and compiled all available (approximately 500 GB) open-source geospatial data (satellite data, reports and maps) relevant to assessing deforestation in the Indonesian Papua and West Papua provinces.
- (ii) Guided the conversion of data into actionable information.

From: March 2019

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

Activities performed:

To: May 2019

Sustainable Highlands Highway Investment Program
RENARDET S.A., rue Rothschild No. 50, Genève, Switzerland.
Department of Works, Goroka, **Papua New Guinea**. Funded by
ADB LOAN 3547, 3548, AND GRANT 0538

GIS Specialist

GIS services and geospatial analyses for Highlands Highway rehabilitation

Worked on location in Goroka, PNG:

- (i) Generated comprehensive maps of population density and distribution, future years population prognoses, elevation, slope, land cover/land use, earth sciences, and synthesized additional spatial information critical to road design and rehabilitation,
- (ii) Analysed catchments upstream of locations where highway crosses streams and rivers, and performed average monthly streamflow calculations at the crossings, and
- (iii) Proposed alternative Goroka bypass alignments.

From: Aug 2018

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: November 2018

Emergency Earthquake Relief Lombok

Self-employed, Lombok, **Indonesia**

People of North Lombok and Provincial Planning Agency
(Bappeda)

Initiative-taker for emergency Lombok earthquake relief actions

Emergency relief and *pro bono* mapping for earthquake relief

Activities performed:

On location in Lombok:

- (i) Being the first to bring medical and other emergency supplies (tents, blankets, water filters, food, medicines, hygiene items, school articles, etc.) with my boat 'Wahoo' to disaster-stricken areas in North and East Lombok
- (ii) *Pro bono* assistance to the provincial planning agency (Bappeda) with mapping services to pinpoint optimum areas for government emergency shelter establishment
- (iii) GPS survey, design, modelling, repair of existing water pipe and installation of a 1.5 km flexible water pipe from high up mountain springs to stricken villages near the coast in Dusun Tangga, Desa Selengen, Kecamatan Kayangan, North Lombok Regency

See: <https://youtu.be/ug0IIRoRD7s>

See also (Dutch) newspaper articles:

<https://www.volkskrant.nl/nieuws-achtergrond/nederlander-vaart-met-hulpgoederen-naar-het-noorden-van-lombok~b65d2714/>

<https://www.volkskrant.nl/nieuws-achtergrond/wijnand-69-helpt-aardbevingsslachtoffers-op-zwaar-getroffen-lombok-mensen-kwamen-naar-me-toe-met-tranen-in-hun-ogen-br~b175c76d/>

From: Jan 2016

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: April 2018

PARTICIPATORY MAPPING AND PLANNING (PMaP) SUB-ACTIVITY: Implementation of PMaP2 and PMaP7 in NTB, NTT, Jambi and Riau provinces

LAND EQUITY INTERNATIONAL Suite 12, 74 Kembla St
WOLLONGONG NSW 2500 PO Box 798

WOLLONGONG NSW 2520 Australia

Millennium Challenge Account – **Indonesia**, Jalan Menteng
Raya, kaveling 21, Jakarta Pusat, Indonesia

Coordinator - GIS Manager geo-spatial databases

Assistance to Indonesian district governments in their spatial planning efforts, through collection and curation of accurate, up-to-date, definitive information into standardized, well-organized geospatial data sets stored in an Information Management System (IMS) and online webGIS through which the data became available to all stakeholders.

Worked in Jakarta, in the provinces, and from home base in Bali:

- (i) Managed the acquisition of geospatial data from national, provincial, district Government agencies and from open-source online global data sets, preparation of geographic information system (GIS) databases, and storage in cloud-based repositories.
- (ii) Made available privately owned LandScan Population Database to the project.
- (iii) Compilation and geo-referencing of existing and pending licences and permits for land and natural resource use, including water use.
- (iv) Collected relevant laws, regulations, decrees and other legal instruments, and conduct an analysis of legal and regulatory procedures for licensing, land administration and natural resources management at the national, provincial, and district levels
- (v) Enhanced district spatial plans through capacity building in spatial planning, enforcement and management of land use information in cloud-based spatially-enabled databases.
- (vi) Managed the development and implementation of webGIS applications allowing government staff to make informed spatial planning decisions.
- (vii) Managed the development and implementation of data import tools allowing spatial planners to import their own data.
- (viii) Managed the development and implementation of apps seeking community engagement and participation regarding spatial planning from the public through online channels.
- (ix) Supported GIS/GPS/remote sensing and webGIS/IMS training activities: Basic GIS Training, Advanced GIS Training, Workshops, and On-The-Job training

From: Oct 2015

To: Feb 2017 (intermittent)

Name of assignment or project:	STATCAP-CERDAS: Statistical Transformation and Quality Improvement, National Bureau of Statistics (BPS), Indonesia
Employer:	Oxford Policy Management Limited, 6 St Aldates Courtyard, 38 St Aldates, Oxford, OX1 1 BN ("OPM"), UK. National Bureau of Statistics, Indonesia /World Bank)
Client:	National Bureau of Statistics, Indonesia , funded by the World Bank.
Positions held:	GIS Advisor
Main Project Features:	The Project aimed to move the BPS statistical programme from a set of semi-autonomous surveys to a harmonised suite of surveys that draw on a common statistical infrastructure. Worked at location in Jakarta, and from home base in Bali:
Activities performed:	(i) Assisted in recognizing common features that underlie all surveys in the form of a Generic Statistical Business Process Model (GSBPM). This was the starting point for harmonisation of operations. (ii) Assisted in building corporate statistical infrastructure required to support as many of the common processes identified in the model as possible and to identify those which will be the mandate of specialist subject matter teams. (iii) Assisted in progressively re-engineering the surveys component and building new application systems based on the corporate statistical infrastructure so that the goals of improved harmonisation, integration and efficiency are achieved, and (iv) Delivered essential GIS and remote sensing expertise in direct support of project objectives.
From: Jul 2015	To: 31 Dec 2015 (intermittent)
Name of assignment or project:	SC 105506 PAK: Punjab Intermediate Cities Improvement Investment Program - Integrated City Development Strategy
Employer:	Asian Development Bank, Manila, Philippines.
Client:	Urban Development and Water Division, CWRD - Central and West Asia Department.
Positions held:	International GIS Specialist
Main Project Features:	Rapid economic, environmental, social and climate change diagnostics for the cities of Sialkot and Sahiwal, Punjab area, Pakistan
Activities performed:	Worked at location in Lahore, Pakistan, and from home base in Bali: (i) Compiled geographic data, field observation, satellite imagery, aerial photographs, and existing maps as well as data from Pakistan government sources (ii) Created new and layered maps (e.g. population density and climate hazards, water availability) using GIS software and related equipment (iii) Analysed and synthesized relationships among varying types of data and information in support of the identifying climate risks, hazards, vulnerabilities, developing adaptation measures, and designing climate resilient urban infrastructure and communities; and (iv) Facilitated the compilation and uploading to a data repository of generated relevant GIS based data and information
From: Nov 2014	To: June 2015
Name of assignment or project:	RIO TINTO SIMANDOU PROJECT- Plan d'Action de Réinstallation et de Compensation
Employer:	URS AUSTRALIA PTY LTD registered office 4/407 Pacific Highway, Artarmon, Australia 2064 ("URS") and Rio Tinto, London, UK.

Client:	SIMFER S.A., Address: Immeuble Bellevue, Boulevard de Bellevue DI 536 BP 848 Conakry, Republic of Guinea – RIO TINTO SIMANDOU PROJECT- Plan d’Action de Réinstallation et de Compensation, funded by Rio Tinto.
Positions held: Main Project Features:	GIS and Image Interpretation Specialist The Simandou project provides access to one of the world’s largest untapped (over 2 billion tonnes), high grade iron ore resources in the world. It can sustain a mine life in excess of 40 years and has the potential to make Guinea one of the world’s top iron ore exporters. The study aimed to specify investment requirements prior to the mine development. Working mainly from home base because EBOLA outbreak prevented fieldwork:
Activities performed:	(i) Setting up of a GIS for land cover mapping along trans-Guinean rail road corridor (ii) Interpretation of recent, high-resolution (50 cm) satellite imagery along a wide swath of the area straddling the trans Guinea railroad alignment (a zone of 650 km by 20 km) (iii) Detailed land use and land cover mapping using mainly Esri® ArcMap and Blue Marble® Global Mapper
From: Jul 2014 Name of assignment or project:	To: June 2015 TA7274 REG: Enabling Climate Change Interventions for Central and West Asia.
Employer: Client:	Asian Development Bank, Manila, Philippines . ADB Natural Resources & Agriculture Division, Central and West Asia Department
Positions held: Main Project Features:	International GIS Expert The overall objective of TA7274 was to mainstream climate change amongst Asian Development Bank (ADB) developing member countries (DMCs) in the Central West Asian region (CWA). Impact was low-carbon inclusive economic growth and climate-resilient development.
Activities performed:	Worked from home base in Bali and at ADB HQ Manila. (i) Conceptualized and managed the building of a state-of-the-art cloud-based GIS system with sophisticated analysis capabilities named ‘Astro’. The Astro GIS facility serves as a reconnaissance tool to assist ADB project officers in project planning and policy making related to land and water management, climate risk screening and greenhouse gas emission analysis in the Central and West Asia region (ii) Sourced and curated spatial data (raster and vector) to be included in the cloud-based GIS analysis tool (iii) Prepared Inception Report, Technical notes, Work Plans, Manual of Operations, Final Report, ADB CWA Spatial Data Repository with Documentation and Users’ Manual, and (iv) Delivered hands-on training for designated ADB staff in the use of Esri® ArcGIS, GIS tools, and to facilitate handover.
From: May 2014 Name of assignment or project: Employer:	To: Jul 2014 Iraq Opportunities Program (Foras) Experience International, Inc. 239/2 Lumpini 1 Building, Suite 42, 4th Floor, Sarasin Road, Lumpini, Pathumwan. Bangkok 10330, Thailand.
Client:	Government of Iraq , funded by USAID
Positions held: Main Project Features:	Senior GIS Specialist A two-year program funded by USAID assisting the Government of Iraq in bringing together job seekers and potential employers using SMS push technology and GIS.
Activities performed:	A follow-up assignment, worked on location in Erbil, and from home base in Bali after repatriation because of ISIS terrorist scare: (i) Continued building a GIS display framework that incorporates project themes and displays changes over time (ii) Identifying base layers required for each project catchment area (iii) Specifying additional layers needed to explain project activities, inputs, and outputs (iv) Training of counterparts (v) Prepared training manuals and technical reports

From: Oct 2013 Name of assignment or project: Employer: Client: Positions held: Main Project Features:	To: Mar 2014 Design of a census-independent sampling strategy for household surveys Sistemas Integrales, José Miguel de la Barra 412 - Casilla 13168 - Santiago, Chile. Development Research Group, World Bank, 1818 H Street NW, Washington, DC 20433, USA GIS-based Sampling Specialist Proposal of an alternative sampling strategy intended to deliver a sample of households with well-defined probabilities, based on the basis of GIS techniques, GPS, high-resolution satellite imagery, and additional information sources other than censuses, in particular the LandScan population database. The standard sampling strategy is critically dependent on reliable and accessible census databases and census cartography. National statistical agencies are the sole source of such data in many countries, which can make them hard to reach. Even when available, census databases are often unreliable and almost always outdated. Activities performed: Worked from home base in Bali on Myanmar population data: (i) Contributed to the formulation of the alternative sampling strategy (ii) Worked on the GIS and hi-res satellite imagery components (iii) Generated all cartographic outputs for the final report
From: Jul 2013 Name of assignment or project: Employer: Client: Positions held: Main Project Features:	To: Aug 2013 Iraq Opportunities Program (Foras) Experience International, Inc. 239/2 Lumpini 1 Building, Suite 42, 4th Floor, Sarasin Road, Lumpini, Pathumwan. Bangkok 10330, Thailand. Government of Iraq , funded by USAID Senior GIS Specialist A two-year program funded by USAID assisting the Government of Iraq in bringing together job seekers and potential employers using SMS push technology and GIS. Worked on location in Erbil, Iraqi Kurdistan: (i) Architected and operationalized a geospatial database and Management Information System (ii) Determining organization-wide GIS needs by performing a User Needs Analysis (iii) Building a GIS display framework that incorporates project themes and displays changes over time (iv) Identifying base layers required for each project catchment area (v) Specifying additional layers needed to explain project activities, inputs and outputs (vi) Training of counterparts (vii) Prepared training manuals, technical reports
From: May 2013 Name of assignment or project: Employer: Client: Positions held: Main Project Features:	To: Dec 2013 Framework contract assisting the National Institute of Statistics of Rwanda in implementing GIS and Mapping Techniques Oxford Policy Management Limited, 6 St Aldates Courtyard, 38 St Aldates, Oxford, OX1 1 BN ("OPM"), UK. National Institute of Statistics of Rwanda (NISR), funded by DFID GIS and Mapping Specialist Responsible for assistance to NISR in mapping and GIS techniques related to the 2012 census Worked onsite in Kigali and Mgahinga area in Rwanda: (i) Completed approximately 80 thematic maps based on the 2012 census of Rwanda (ii) Designed and implemented a Dashboard for easy viewing of thematic maps (iii) Upskilled and strengthened capacity of NISR GIS staff and (iv) Prepared training manuals
From: Feb 2013	To: Feb 2013

Name of assignment or project: Employer: Client: Positions held: Main Project Features: Activities performed:	Myanmar National Spatial Data Infrastructure UNDP, UN Building, 6 Natmauk Road, Yangon. Myanmar Information Management Unit (MIMU), funded by UNDP Spatial Data Specialist Assisting MIMU with formulating a draft Myanmar National Spatial Data Infrastructure (NSDI) Worked in Yangon and Naypyidaw, Myanmar: (i) Met with stakeholders (institutions using or producing spatial data), and (ii) Formulated a preliminary plan for the setting up of an NSDI
From: Oct 2012 Name of assignment or project: Employer: Client: Positions held: Main Project Features: Activities performed:	To: Nov 2012 Assistance to the National Institute of Statistics (INE) Oxford Policy Management Limited, 6 St Aldates Courtyard, 38 St Aldates, Oxford, OX1 1 BN ("OPM"), UK. National Institute of Statistics (INE), Maputo, Mozambique funded by DFID Database and Mapping Adviser Delivering targeted GIS and mapping capacity building to selected INE staff, focusing on techniques to visualize and communicate the statistical databases. A follow-up assignment, working in Maputo: (i) Producing animated maps showing key statistical indicators. (ii) Strengthened INE's capacity to produce such maps independently. (iii) Introducing ways to use Google Earth imagery effectively. (iv) Presented findings and insights to senior management, and (v) Wrote technical reports.
From: May 2012 Name of assignment or project: Employer: Client: Positions held: Main Project Features: Activities performed:	To: Nov 2012 Myanmar 2014 Population and Housing Census UNFPA, UN Building, 6 Natmauk Road, Yangon, Myanmar. Ministry of Immigration & Population, Myanmar, Department of Population (DOP) - funded by UNFPA Census Cartography Adviser Assess the census cartography plan for the Myanmar 2014 Census, propose new methodologies and provide technical assistance Working in Yangon and Nay Pyi Taw: (i) Designed an appropriate GIS system for use by DOP, (ii) Designed the spatial database, (iii) Specified, verified and installed computer hardware, software and peripheral equipment, (iv) Delivered hands-on technical assistance in census cartography, GPS technology, and remote sensing methodologies to 20 DOP staff, (v) Delivered a dedicated management training course to high-level Ministry staff in GIS comprehension (what GIS can - and cannot - accomplish), (vi) Generated a country-wide geo-referenced mosaic of scanned 50k and 250k scale topographic maps serving as the national mapping base and backdrop, and (vii) Mapped and coded township maps to serve as starting point for census mapping.
From: Dec 2011 Name of assignment or project: Employer: Client: Positions held: Main Project Features:	To: Dec 2012 Iraqi Kurdistan Christianity Project Michael-Moran Associates LLC, 3705 Taylor Street, Chevy Chase, MD 20815, USA Iraqi Kurdistan Regional Government, funded by KRG Mapping and Surveying Specialist, Survey Director A public relations project with Catholic University (Washington DC) for Iraqi Kurdistan to demonstrate and quantify the level of religious freedom there. Target audiences were U.S. Christians and the U.S. Congress.

Activities performed:

Working in Iraqi Kurdistan:

- (i) Production of the first-ever Iraqi Kurdistan Christianity Map and published online,
- (ii) Designing and implementing a random sample survey of the Christian population of a Christian enclave (Ainkawa near Erbil),
- (iii) Interviewed and videoed numerous Christian inhabitants throughout Iraqi Kurdistan, and
- (iv) Presented map and survey results during a conference in Washington DC (December 2012).

From: Sep 2010**Name of assignment or project:****Employer:****Client:****Positions held:****Main Project Features:****To:** Aug 2011 (Intermittent)

Enabling Climate Change Interventions in Central and West Asia: CWA Hazard and Risk Map, Wind and Solar Energy Atlases - TA 7274-REG)

Cardno Emerging Markets (UK) Ltd/ICF International, Asian Development Bank (ADB), Manila, **Philippines, GIS Modelling and Mapping Specialist**

TA to advise the Bank in pinpointing optimum locations for investment in renewable energy power-generating plants in ten Central and West Asian countries (Afghanistan, Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Pakistan, Tajikistan, Turkmenistan, Uzbekistan)

Working mainly from home:

Activities performed:

- (i) Designed and operationalized a geospatial database enabling mapping of the ten CWA countries via the Geospatial Toolkit, an NREL-developed map-based software application,
- (ii) Determined sub-regional distributions of solar and wind resources to a high spatial resolution (2 arc-minutes),
- (iii) Put together a highly visual, region-wide, country level, and sub-regional color-coded GIS-based (and hard copy) solar and wind resource atlas,
- (iv) Mapped out solar and wind power plant location constraints and interfaced with the solar and wind atlases,
- (v) Pioneered and produced 4-D maps i.e. moving map displays (loops) to depict climate changes over time (seasonal and era) and investment scenarios depicting rates of return areas with varying prime rates (both "space-time" and "space-economic parameter data-cubes") of the ten Central and West Asian countries showing optimum locations for renewable energy power plants, and
- (vi) Presented results at ADB HQ, Manila.

From: Mar 2011**Name of assignment or project:****Employer:****Client:****Positions held:****Main Project Features:****To:** Apr 2011National Risk and Vulnerability Assessment ICON-INSTITUT Public Sector GmbH, Cologne, Germany Central Statistical Organization (CSO), Kabul, **Afghanistan**, funded by European Union, with GIS component separately sponsored by DFID, Kabul**GIS and Mapping Specialist**

Technical Assistance, GIS and Mapping Support to CSO for conducting the National Risk and Vulnerability Assessment

Working in Kabul:**Activities performed:**

- (i) Delivered advanced mapping technique training to 20 CSO GIS Unit staff,
- (ii) Introduced the combination of SRTM shaded relief anaglyphs with Russian 1/50k topo maps,
- (iii) Equipped 30 Afghan NRVA survey supervisors with skills in the use of hand-held GPS,
- (iv) Demonstrated techniques for effective GPS/GIS integration to CSO GIS Unit staff,
- (v) Architected and implemented a spatial database structure for the CSO GIS Unit, and
- (vi) Produced, in collaboration with assigned CSO GIS Unit staff, highly sophisticated NRVA survey map templates.

From: Dec 2010**Name of assignment or project:****To:** Dec 2010

Assistance to National Institute of Statistics (INE)

Employer: Client: Positions held: Main Project Features: Activities performed:	Oxford Policy Management Limited, 6 St Aldates Courtyard, 38 St Aldates, Oxford, OX1 1 BN ("OPM"). National Institute of Statistics (INE) in Mozambique Database and Mapping Advisor Delivering targeted GIS and mapping capacity building to selected INE staff, focusing on advanced techniques to visualize and communicate statistical databases Follow-up mission, working in Maputo: (i) Assisted in setting up the INE geospatial database, (ii) Producing animated maps showing key statistical indicators, (iii) Strengthened INE's capacity to produce such maps independently, (iv) Compression and mosaicking of topo maps to alleviate pressure on server and for use as thematic map backdrop, (v) Introducing ways to use Google Earth imagery effectively, (vi) Presented findings and insights to senior management, and (vii) Wrote technical reports.
From: Apr 2010 Name of assignment or project: Employer: Client: Positions held: Main Project Features: Activities performed:	To: June 2010 Optimum future school location assessment Maxwell Stamp PLC, Abbot's Court 34 Farringdon Lane London EC1R 3AX United Kingdom Abu Dhabi Education Council – ADEC, funded by DFID Spatial Planning and GIS Expert Future school location planning Applied advanced GIS spatial modelling to determine optimum future school locations
From: Mar 2010 Name of assignment or project: Employer: Client: Positions held: Main Project Features: Activities performed:	To: Apr 2010 Training in sampling techniques to Direction Générale des Statistiques du Gabon The World Bank, Libreville, Gabon Direction Générale des Statistiques (DGS) du Gabon, funded by the World Bank Expert en Techniques des Sondages Delivery of a training course in sampling techniques to designated DGS staff Working in Libreville, Gabon : (i) Trained designated staff of the statistics office (DGS) in sampling techniques applied to land use surveys involving random survey starting points, (ii) Introduced software to quickly calculate margins of error with varying sampling, cluster and population sizes, confidence interval and prevalence, (iii) Delivered basic training in the use of handheld GPS and GIS related to random sampling, and (iv) Performed field exercises in Libreville taking random household samples. Training was delivered as the sole lecturer, and entirely in French language
From: Sep 2008 Name of assignment or project: Employer: Client: Positions held: Main Project Features:	To: Dec 2009 (Intermittent) GDP Assessment of Iraqi Kurdistan Michael-Moran Associates LLC, 3705 Taylor Street, Chevy Chase, MD 20815, USA Iraqi Kurdistan Regional Government, Kurdistan Region Statistical Office (KRSO), Ministry of Planning GIS, GPS and Mapping Adviser, Survey Manager Delivering GIS, GPS and mapping services for Industrial Output surveys instrumental in determining the GDP of the Region

Activities performed:

Working in Iraqi Kurdistan:

- (i) Determined the statistics office (KRSO) GIS requirements and designed an appropriate GIS system,
- (ii) Upskilled designated KRSO staff in GIS and GPS techniques applied to surveys involving random survey starting points,
- (iii) Delivering GIS, GPS and mapping services for Industrial Output surveys critical to determining the GDP of the Region,
- (iv) Played key role in improving information management,
- (v) Managing fieldwork operations for region-wide Industrial Output surveys,
- (vi) Spatial analysis of survey data to determine expansion factors, and
- (vii) Managing finances of the GDP project.

From: Oct 2009

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Nov 2009

Assistance to National Institute of Statistics (INE)
Oxford Policy Management Limited, 6 St Aldates Courtyard,
38 St Aldates, Oxford, OX1 1 BN ("OPM").

National Institute of Statistics (INE) in **Mozambique**

Database and Mapping Advisor

Delivering targeted GIS and mapping capacity building to selected INE staff, focusing on advanced techniques to visualize and communicate statistical databases

First visit, working in Maputo:

- (i) Assisted in setting up the INE geospatial database,
- (ii) Producing animated maps showing key statistical indicators,
- (iii) Strengthened INE's capacity to produce such maps independently, and
- (iv) Wrote technical reports.

From: Mar 2009

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Apr 2009

Technical Assistance for Natural Disaster Risk Management
Ministry of Agriculture and Rural Development (MARD)

Viet Nam

Department of Dyke Management and Flood and Storm Control (DDMFSC) and Central Project Office (CPO), funded by AusAID

GIS Specialist (Disaster Management)

Consulting services and equipment (hard/software) for upgrading GIS facilities and capabilities for the application of GIS to disaster risk management at the central and regional levels.

Working in Viet Nam:

- (i) Preparation of Terms of Reference for a GIS software and hardware and GPS package, and
- (ii) Field visits to Da Nang and Ho Chi Min City

From: Oct 2008

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Feb 2009

Vanuatu 2009 National Population and Housing Census
Secretariat of Pacific Communities (SPC)

Vanuatu National Bureau of Statistics

GIS/GPS Consultant for Census

In preparation for the 2009 census, provide GPS for census expertise earlier obtained in Timor-Leste

Worked in Vanuatu and from home:

- (i) Assisted with mapping of enumeration areas, and
- (ii) Trained staff in the use of handheld GPS.

From: Jul 2008

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Dec 2008 (Intermittent)

Mongolia 2010 Population and Housing Census

UNFPA, Ulaan Baatar, **Mongolia**

Census Bureau in the National Statistical Office (NSO) of Mongolia, funded by UNDP

GIS and GPS Advisor for Census

Introduction of GIS, GPS and Mapping for the Population and Housing Census

Activities performed:

Working in Ulaan Baatar, Mongolia:

- (i) Developed a plan for use of GIS and GPS in the 2010 Population and Housing Census
- (ii) Converted NSO data into thematic map information
- (iii) Delivered basic and advanced GIS and GPS training to selected NSO staff, with emphasis on spatial analysis and population modelling using GPS household location waypoints.
- (iv) Training manual writing and final reporting

From: Jun 2007

Name of assignment or project:
Employer:

Client:

Positions held:

Main Project Features:

Activities performed:

To: Jun 2008 (Intermittent)

GIS assistance to Ministry of Municipalities
Michael-Moran Associates LLC, 3705 Taylor Street, Chevy Chase, MD 20815, USA

Iraqi Kurdistan Regional Government - KRG, Ministry of Municipalities

GIS, GPS and Mapping Adviser

Assessment of Kurdish inhabitants in Kirkuk prior to Baghdad-controlled census

Working in Erbil, Iraqi Kurdistan:

- (i) Rendered operational the GIS Unit of Iraqi Kurdistan Regional Government Ministry of Municipalities through pertinent technical skill transfers,
- (ii) Created demand for GIS services within Urban Planning Department,
- (iii) Together with GIS staff devised and published a new methodology of village boundary mapping involving scanned and georeferenced schematic maps and SRTM shaded relief data, and
- (iv) Delivered targeted GIS and GPS training at the Ministry of Municipalities, with emphasis on
 - a) GPS procedures,
 - b) conversion into GIS format,
 - c) linking of spatial location (GPS) database with 'content' (questionnaire) database, and
 - d) spatial analysis and modelling

From: Jul 2007

Name of assignment or project:
Employer:

Client:

Positions held:

Main Project Features:

Activities performed:

To: Jul 2007

Abu Dhabi Spatial Data Infrastructure (SDI)
Michael-Moran Associates LLC,
3705 Taylor Street, Chevy Chase, MD 20815, USA. Seconded to Geographic Planning Collaborative, Inc. (GPC)

Abu Dhabi Systems & Information Council - ADSIC)

GIS and GPS Adviser for Census

In the framework of the establishment of the Abu Dhabi Spatial Data Infrastructure (SDI), was charged with examining and evaluating the spatial aspects of the country's census data.

Working in Abu Dhabi:

- (i) Conversion of household locations obtained with handheld GPS during census into a GIS format
- (ii) Linking of population and housing census enumeration data with GPS household locations
- (iii) Performing spatial analysis and mapping of various demographic aspects derived from the combined GPS and census data sets, and
- (iv) Presented geospatial analysis results to senior representatives of all agencies with a stake in the national SDI

From: Oct 2006

Name of assignment or project:
Employer:

Client:

Positions held:

Main Project Features:

To: Mar 2007 (Intermittent)

Vulnerability, Analysis and Mapping (VAM)
World Food Programme (WFP), Dili, **Timor-Leste**
WFP, Dili

GIS and Mapping Advisor

To assist WFP Dili with GIS capability for Vulnerability, Analysis and Mapping (VAM) activities.

Activities performed:

Working in East Timor:

- (i) Assessment of present situation and capabilities in terms of human resources, mapping equipment
- (ii) Training of local staff in general GIS principles and data collection, and
- (iii) Played key role in designing and implementing of thematic map series depicting a series of vulnerabilities per sub district

From: Nov 2006

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Apr 2008 (Intermittent)

2007 Fiji Islands Population Census

UNFPA Suva

Fiji Islands Bureau of Statistics, funded by United Nations Population Fund

GIS and GPS Adviser for Census

To assist the Fiji Islands Bureau of Statistics with implementing GPS and GIS techniques, and spatial analysis for the 2007 Fiji Islands Population Census.

Working in Suva and in-country:

- (i) Assessment of present situation and capabilities in terms of human resources, equipment, mapping, and training of enumerators
- (ii) Assisting with census enumeration area mapping efforts, and provide GIS and GPS on-the-job training
- (iii) Setting up of training modules for both GPS operators (taking waypoints) and supervisors (downloading waypoints)
- (iv) Assisting with GPS waypoint downloading, mapping and management aspects of the October 2006 Pilot Census and the 2007 Census, and
- (v) Assisting with post Census waypoint data processing, spatial analysis, spatial modelling, and mapping of results

From: Sep 2006

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Mar 2007

Post enumeration survey of Kirkuk

Michael-Moran Associates LLC, 3705 Taylor Street, Chevy Chase, MD 20815, USA

Iraqi Kurdistan Regional Government, Ministry of Planning, funded by KRG

GIS, GPS and Mapping Adviser

Charged with designing and conducting a post enumeration survey, and capacity building of KRG Statistics Office staff Working in Erbil and in-country:

- (i) Designed a method to sample a census using random point generation in a GIS environment.
- (ii) Constructed a sampling frame based on spatially random points
- (iii) Assisted with the implementation of GPS fieldwork operations
- (iv) Was in charge of the spatial GIS component of the post enumeration analysis
- (v) Produced authoritative maps for the post-enumeration validation report, and
- (vi) Delivered targeted GIS and GPS training at the Ministry of Municipalities, covering (i) GPS procedures, (ii) conversion into GIS format, (iii) linking of spatial location (GPS) database with 'content' (questionnaire) database, and (iv) spatial analysis and modelling

From: Aug 2006

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

To: Oct 2007 (Intermittent)

Environmental Impact Assessment (EIA) studies for Iralalaru Hydropower and Lospalos-Dili Transmission Line Projects EPANZ Services P/L, P.O. Box 268 Mangonui, 0442, New Zealand.

Government of **Timor-Leste**, Department of Environment, funded by Norconsult, Norway

GIS and Mapping Specialist

To provide mapping services for Environmental Impact Assessment (EIA) studies for Iralalaru Hydropower and Lospalos-Dili Transmission Line Projects.

Activities performed:

Working in-country in Timor-Leste and from home:

- (i) Preparation of maps and GPS units in preparation of field visits by EIA team members,
- (ii) Production of a series of maps and publication-grade layout designs for themes relevant to Lospalos-Dili power line and Iralalaru hydropower EIA studies

From: Sep 2006

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

Activities performed:

To: Sep 2006

GPS use in household and population surveys

Michael-Moran Associates LLC, 3705 Taylor Street, Chevy Chase, MD 20815, USA

Iraqi Kurdistan Regional Government, Kurdistan Regional Statistical Office (KRSO), funded by the Prime Minister's Office

GIS and GPS Advisor for Census

To provide the Kurdistan Regional Statistical Office (KRSO) with specific expertise related to the use of GPS for household surveys and census taking.

Worked in Erbil, Iraqi Kurdistan:

- (i) Merging of GPS waypoint files of the Kirkuk pre-census survey into a household location database
- (ii) Editing of GPS data
- (iii) Joining of household location database with questionnaire content database, and
- (iv) Capacity building of Kurdistan regional statistics office (KRSO) staff

From: Feb 2005

Name of assignment or project:

Employer:

Client:

Positions held:

Main Project Features:

Activities performed:

To: Jul 2006

Mapping Assistance to the National Directorate for Land and Property

Illawara Technology Corporation Limited, Australia

Government of **Timor-Leste**, funded by AusAID

National Mapping Adviser

As National Mapping Adviser charged with assisting the National Directorate for Land and Property (DNTP) in the coordination of national mapping.

Working in Dili and in-country:

- (i) Capacity building of GIS and mapping staff at DNTP:
Delivered formal training in the following subjects:
 - a. general GIS, GPS/GIS integration,
 - b. map adjustment techniques,
 - c. CAD-GIS conversions,
 - d. spatial analysis,
 - e. raster image processing and integration with GPS/GIS,
 - f. inter-agency upgrade to ArcGIS 9
- (ii) Established and chaired a national GIS user group serving as an inter-agency platform for data sharing and best-practice exchange,
- (iii) Assessed and upgraded base maps, and
- (iv) Established the first draft of a Timor-Leste national Spatial Data Infrastructure (SDI), involving developing of policies for updating, exchange and integration of spatial data, and setting standards for cartographic output

From: May 2003
Name of assignment or project:
Employer:
Client:

Positions held:
Main Project Features:

Activities performed:

To: Jan 2005
Timor-Leste 2004 Population and Housing Census
United Nations Population Fund (UNFPA), Dili
Timor-Leste Director General of Statistics, Ministry of Finance,
funded by UNFPA

Census Mapping Manager

First census taking since many decades. The majority of households in Timor-Leste have no address, so it was decided to use GPS for the geo-location of all households. Was the first to develop and implement a comprehensive GPS-based system to create a national Census household location database.

- (i) Trained counterpart staff at National Statistics Directorate in the use of GPS and GIS for census purposes,
- (ii) Generated approximately 2,000 Enumeration Area maps for the 2004 Census,
- (iii) Pioneered the use of GPS for census,
- (iv) Directed and managed a nationwide comprehensive GPS survey of all households,
- (v) Purchased 730 handheld GPS units,
- (vi) Delivered enumerator training,
- (vii) Linked GPS waypoints with census questionnaire database,
- (viii) Generated thematic maps for each census topic, and
- (ix) Through spatial analysis and modelling, established correlations and causalities from the combination of Census results with 3rd party spatial information

From: Oct 2003
Name of assignment or project:
Employer:
Client:

Positions held:
Main Project Features:
Activities performed:

To: Nov 2003
East Timor Land Law Program
ARD, Inc., Burlington Vermont, USA
Timor-Leste National Directorate for Land and Property,
funded by USAID

GIS and GPS Consultant

Capacity building of university staff
Worked in Dili:

- (i) Delivered GIS and GPS training in Indonesian to University of Timor-Leste lecturers and designated ARD research staff
- (ii) Generated reference maps for use by research staff in conjunction with GPS

From: Dec 2002
Name of assignment or project:
Employer:
Client:
Positions held:
Main Project Features:

Activities performed:

To: Apr 2003
Mapping and planning of health facilities
World Health Organization (WHO), Dili, Timor-Leste
Ministry of Health, Dili, **Timor-Leste**, funded by WHO
GIS Specialist

Setting up a GIS for planning, implementation and management of health facilities

Working in Dili and in-country:

- (i) Has set up a GIS for planning, implementation and management of health facilities
- (ii) Delivered multilingual GIS, Image Analysis and GPS training to WHO staff (English) and Ministry of Health counterparts (Bahasa Indonesia)
- (iii) Performed mapping of disease incidence at sub-district level
- (iv) Mapped all health facilities country-wide and set up a spatial database still used today
- (v) Performed spatial analyses to facilitate outbreak predictions, and
- (vi) Modelled population data combined with health facility data to arrive at optimum health facility locations taking into account travel distances

From: Nov 2002
Name of assignment or project:
Employer:
Client:
Positions held:
Main Project Features:
Activities performed:

To: Dec 2002
Capacity building at Mineral Resources
Department of Energy and Mineral Resources, **Timor-Leste**
Idem, funded by USAID
GIS and Mapping Specialist
Capacity building of Mineral Resources staff
Working in Dili:
(i) GIS and GPS training in Bahasa Indonesia to 8 staff of Mineral Resources,
(ii) Designed, produced and operationalized a Mineral Rights Administration and Information System: a GIS application to be used by the Department to track mineral exploration claims

From: Jun 2002
Name of assignment or project:
Employer:
Client:

To: Oct 2002
Agriculture and Land Use Mapping and GIS Training (extension)
AgWest International, Agriculture Western Australia, Perth, Australia
Government of **Timor-Leste**, Ministry of Agriculture, Forestry and Fisheries (MAFF), funded by AusAID

Positions held:
Main Project Features:

GIS Data Management Specialist
The project extension aimed to assist MAFF to develop sustainable capacity to manage its GIS.

Activities performed:

Worked in Dili:
(i) Overall management of the GIS database
(ii) All data capture for the GIS database
(iii) Formulation of metadata statements describing the 500+ geo-datasets of the GIS unit
(iv) Writing of manuals and provision of training to GIS staff in air photo and satellite data interpretation
(v) Creation of a unique climate geo-dataset based on historic climatic databases
(vi) Spatial modelling of land suitability analysis for 13 land use types based on climate and physical land characteristics, and produced potential land suitability maps for these LUTs
(vii) Writing of manuals and provision of advanced training to MAFF selected staff in the use of image analysis software programs, and
(viii) Writing of manuals and provision of training to MAFF's management on awareness about nature and capability of GIS

From: Apr 2002
Name of assignment or project:
Employer:
Client:

To: May 2002
Mahakam Delta Stakeholder Assessment Project
PT Win, Jakarta, Indonesia
TotalFinaElf E&P Indonésie, Balikpapan, Kalimantan Timur, **Indonesia**

Positions held:
Main Project Features:
Activities performed:

GIS and Land Use Mapping Specialist
Mapping of historic land use in the Mahakam Delta
Working in Samarinda:
(i) Creation of time series of land cover maps in the Mahakam Delta, to quantify changes in the mangrove environment over the last three decades, and
(ii) Assisting TotalFinaElf in aligning existing maps, produced by the environmental division, with the base map from their topographic division.

From: Feb 2002
Name of assignment or project:
Employer:
Client:
Positions held:
Main Project Features:

Activities performed:

To: Mar 2002
Natural disaster preparedness enhancement
UNDP Dili, Timor-Leste
National Disaster Management Office, **Timor-Leste**
GIS and Hazard Mapping Specialist
Aim was to set up and develop a GIS database to assist natural disaster preparedness, response, and recovery & reduction activities.
(i) Training of Timor-Leste staff in Esri ArcView and GPS use
(ii) Setting up of a Disaster Management Information System (hardware, software, data)
(iii) Preparation of a nation-wide Natural Hazard Vulnerability, and Risk Map (8 themes) using existing data, and by generating new information from satellite image data and field verification by UN helicopter, and
(iv) Presentation of results to Timor-Leste Government.

From: Apr 2001
Name of assignment or project:
Employer:
Client:
Positions held:
Main Project Features:

Activities performed:

To: Nov 2001
Agriculture and Land Use Mapping and GIS Training
AgWest International, Agriculture Western Australia, Perth
Government of **Timor-Leste**, Ministry of Agriculture, Forestry and Fisheries (MAFF), funded by AusAID
Computer Data Management Specialist
The project aimed to set up a geographical database to assist in informed agricultural and land use planning decisions, and to provide in-depth training in GIS-related subjects to ensure sustainability of MAFF's GIS Unit.
Working in Dili:
(i) Thematic coverage and database preparation
(ii) Cooperation with other GIS agencies to prevent duplication and to foster standardization
(iii) Preparation of village-level administration maps
(iv) Air photo and satellite data interpretation, and
(v) GIS and database training to East Timorese staff

13. Important projects more than 25 years back

From: Apr 1998
Name of assignment or project:
Employer:
Client:
Positions held:
Main Project Features:

Activities performed:

To: Mar 2001
Satellite Assessment of Rice in Indonesia (ALA/96/01)
Bureau Central d'Études d'Outre-Mer (BCEOM)
Ministry of Agriculture, Republic of **Indonesia**, funded by the European Union
Project Manager (Consortium); Co-Director (Europe)
The project aimed to introduce (and render operational) the use of space-borne remote sensing data (both microwave and optical), with agro-statistical techniques and numerical crop growth modelling in a GIS environment, for nationwide rice monitoring and production forecasting.
Working in Jakarta and in JRC Ispra Italy (two months):
(i) Heading and coordinating a consortium of European companies (French, Dutch, Italian)
(ii) Liaison with Indonesian Project partners and with European Union Representation
(iii) Liaison with European remote sensing, GIS, and agricultural research institutes to ensure a two-way flow of information related to the Project
(iv) Preparation of budgets and work plans
(v) Ensuring timely and correct implementation and execution of the Project in accordance with work plans
(vi) Establishment and implementation of appropriate accounting and financial management procedures
(vii) Preparation of financial accounts and progress reports

From: May 1997
Name of assignment or project:
Employer:
Client:

Positions held:
Main Project Features:

Activities performed:

To: Mar 1998
Marine Resource Evaluation and Planning Project - Part A
Bureau Central d'Études d'Outre-Mer (BCEOM)
Directorate General of Regional Development, Republic
of **Indonesia**, funded by Asian Development Bank (ADB Loan
1203-INO)

Team Leader

Strengthening of Indonesia's marine and coastal planning and
management.

Working in Jakarta:

- (i) Overall project coordination of various long & short term
international and local experts,
- (ii) Liaison with government authorities at national and
provincial level, and Asian Development Bank, and
- (iii) Financial management of both foreign currency and
Rupiah portions of the contract, and
- (iv) Preparation of monthly and quarterly progress reports.

From: Jul 1993
Name of assignment or project:

Employer:
Client:

Positions held:
Main Project Features:

Activities performed:

To: Jan 1997
Second Land Resource Evaluation and Planning Project - Part
D
U L G Consultants Ltd. Wolverhampton, UK
Government of **Indonesia**, Directorate General of Regional
Development, funded by Asian Development Bank (Loan 1099
- INO)

**Regional Team Leader & Physical Planner / GIS
Specialist**

Strengthening of Provincial Physical Planning Capabilities in
East Java, Bali, West and East Nusa Tenggara, and **East
Timor** Provinces by assisting Provincial Planning Bureaus
(Bappeda) in setting-up of a framework for rational planning
by establishing land zones based mainly on land suitability
Based in Bali, working in the provinces:

- (i) Development of an efficient and easy-to-use procedure of
digitizing and geo-coding for Esri ARC/INFO, using
AutoDesk AutoCAD, simplifying the GIS process for staff
with low exposure to Esri ARC/INFO,
- (ii) Development of a technique to extract high-detail Land
Facet information from RePPPOT Land Systems through
combination with slope class information. This technique
allows determination of high-detail land suitability in
uncharted areas,
- (iii) Supervision and on-the-job training of counterparts in GIS
and the production of topographic and thematic maps for
land use, environmental monitoring, etc.,
- (iv) Supervision on data consistency, field verification of
findings,
- (v) Overall regional project coordination of various long &
short term international and local experts,
- (vi) Liaison with government authorities at national, regional
and provincial level, and
- (vii) Preparation of monthly, quarterly progress reports,
progress-briefings.

From: Sep 1989
Name of assignment or project:
Clients:

Positions held:
Main company features:

To: Oct 1992
Co-founding and running of the offshore company QualiTech
Consultants Ltd., Limassol, **Cyprus**

Food and Agriculture Organisation (FAO), International Fund for
Agricultural Development (IFAD), Syrian national mapping
agency, international consulting companies (Euroconsult,
MDF, DHV, BCEOM), Municipality of Makkah, Saudi Arabia,
Ministry of Natural Resources in Cyprus, and a number of
universities in Germany

Managing Director

The company and its staff were situated in Limassol Cyprus,
and specialized in installation and maintenance of digital
image processing and GIS systems at a number of clients'
sites, and providing consultancy services and training in
remote sensing and GIS applications, image processing and
GIS software, and satellite imagery sales. QualiTech was
International EOSAT and TerraMar Representative.

From: Nov 1983
Name of assignment or project:
Employer:
Client:

Positions held:
Main Project Features:

Activities performed:

To: May 1986
Merauke III Transmigration Project
Euroconsult, Arnhem, The Netherlands

Government of **Indonesia**, Directorate General of Regional Development, funded by The World Bank

Soils and Land Evaluation Specialist

The project's goal was to find areas suitable for transmigration in the regions surrounding Merauke, southeast Papua province (formerly Irian Jaya)

Stationed and working in Merauke:

- (i) Production of land unit base maps, land suitability and land use maps and soil maps through air photo interpretation,
- (ii) Interpretation of aerial photographs of vast areas (larger than UK) of upland in South-East Irian Jaya
- (iii) Performing field verification and soil surveying field-work,
- (iv) Design of a physiographic legend which was adopted project-wide , and
- (v) Development of a cartographic method based on Landsat images and aerial photographs to produce high-precision base maps as an alternative to the Slotted Template method

From: Nov 1981
Name of assignment or project:
Employer:
Client:

Positions held:
Main Project Features:

Activities performed:

To: Jul 1983
Nationwide Survey of Tidal Lands
Euroconsult, Jakarta Indonesia

Government of **Indonesia**, Directorate General of Regional Development, funded by Asian Development Bank (Loan 1099 - INO)

Remote Sensing Specialist

Make an inventory of areas suitable for transmigration in the tidal swamp lands of Papua (formerly Irian Jaya), Kalimantan and Sumatra

Based in Jakarta, worked in the provinces:

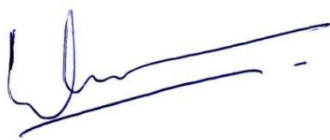
- (i) Interpretation and ground-truthing of all Landsat images, side-looking radar images and aerial photographs of coastal and semi-coastal swamp areas with a total area of about 10 million ha (approx. size of UK),
- (ii) Established an innovative low-cost image preparation laboratory based on Diazo and photographic techniques for production of field-ready imagery for navigation and interpretation,
- (iii) Coordination with local institutions for digital analysis of Landsat data,
- (iv) Field work in Irian Jaya, Kalimantan and Sumatra tidal swamp lands, and
- (v) Training of Indonesian counterparts in image interpretation techniques

From: Sep 1977
Name of assignment or project:
Employer:
Clients:
Positions held:
Activities performed:

To: Oct 1981
Associate Expert position at the Remote Sensing Unit at FAO, Rome, funded by the Dutch Government
Food and Agriculture Organization (FAO) of the United Nations, Rome, **Italy**
Various remote sensing institutes worldwide
Remote Sensing Specialist
Based in Rome, and did missions in various countries:
(i) Missions in Senegal, Bangladesh, China, Sudan
(ii) In China, setting-up two Remote Sensing Laboratories, one at the Beijing Agricultural University and one at the Ministry of Forestry
(iii) In Sudan, mapping of inundation areas near Jonglei, using visual interpretation of Landsat imagery time series,
(iv) In Rome organized and conducted international training courses in cooperation with ESA, United Nations, World Meteorological Organization, and Italian Government in remote sensing use applied to agricultural land suitability assessment and forest inventory,
(v) Developed and fine-tuned production of Diazo satellite image colour composites using densitometer
(vi) Involved in setting up of a database for world-wide imagery, a precursor of a GIS database

14. Certification:

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience. I understand that any wilful misstatement described herein may lead to my disqualification or dismissal, if engaged.



Wijnand Durk LANGERAAR

Signature of staff member or authorized representative of the staff

Date: _____



UNIVERSITAS RHENOTRAIECTINA

L.S.

FACULTAS DISCIPLINARUM MATHEMATICARUM ET PHYSICARUM

Iuveni ornatissimo *Wijnand Durk Langeraar*

nato die *24* mensis *augusti* anno *1948* in urbe *s. Gravenhage*

EXAMINE DOCTORALI rite absoluto *geologiae*
cuius examinis partem principalem obtinebat

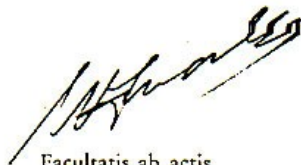
sedimentologia
partes secundarias autem
geologia structuralis et usui applicata
ius metallicum

ius summos petendi in hac Facultate honores concedit;

idque propter egregia eius merita CUM LAUDE.

Datum Traiecti ad Rhenum

die *26* mensis *septembris* anno *1977*



Facultatis ab actis



Facultatis decanus

Ex albo Facultatis no. *0322*



Universitas Rheno-Traiectina

Facultas Disciplinarum

Gebaarmenthe copie